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University of Bath & Diamond Light Source

# COARSE GRAINING AND REFLECTIVITY: A LOVE STORY?

Comp Chem Seminar | 2017-03-23

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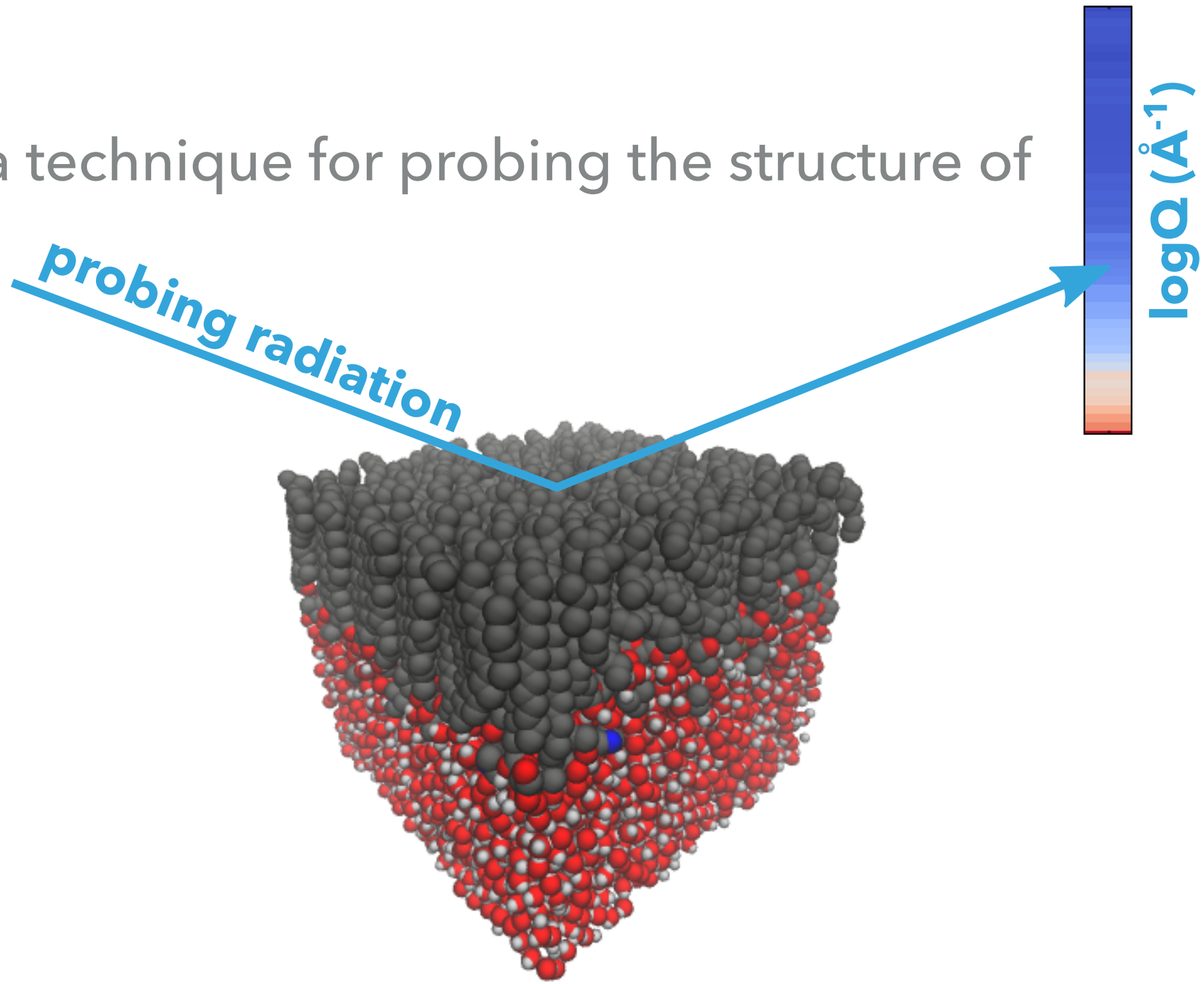
# REFLECTIVITY

- Reflectivity is a technique for probing the structure of interfaces



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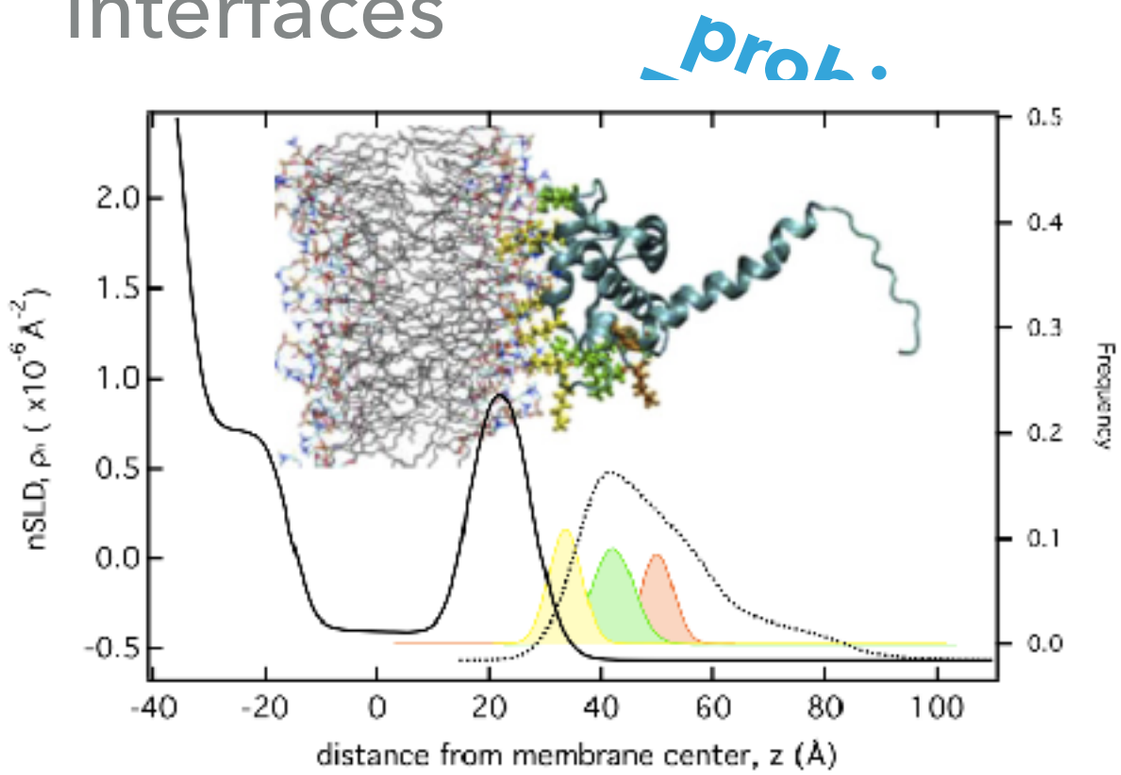
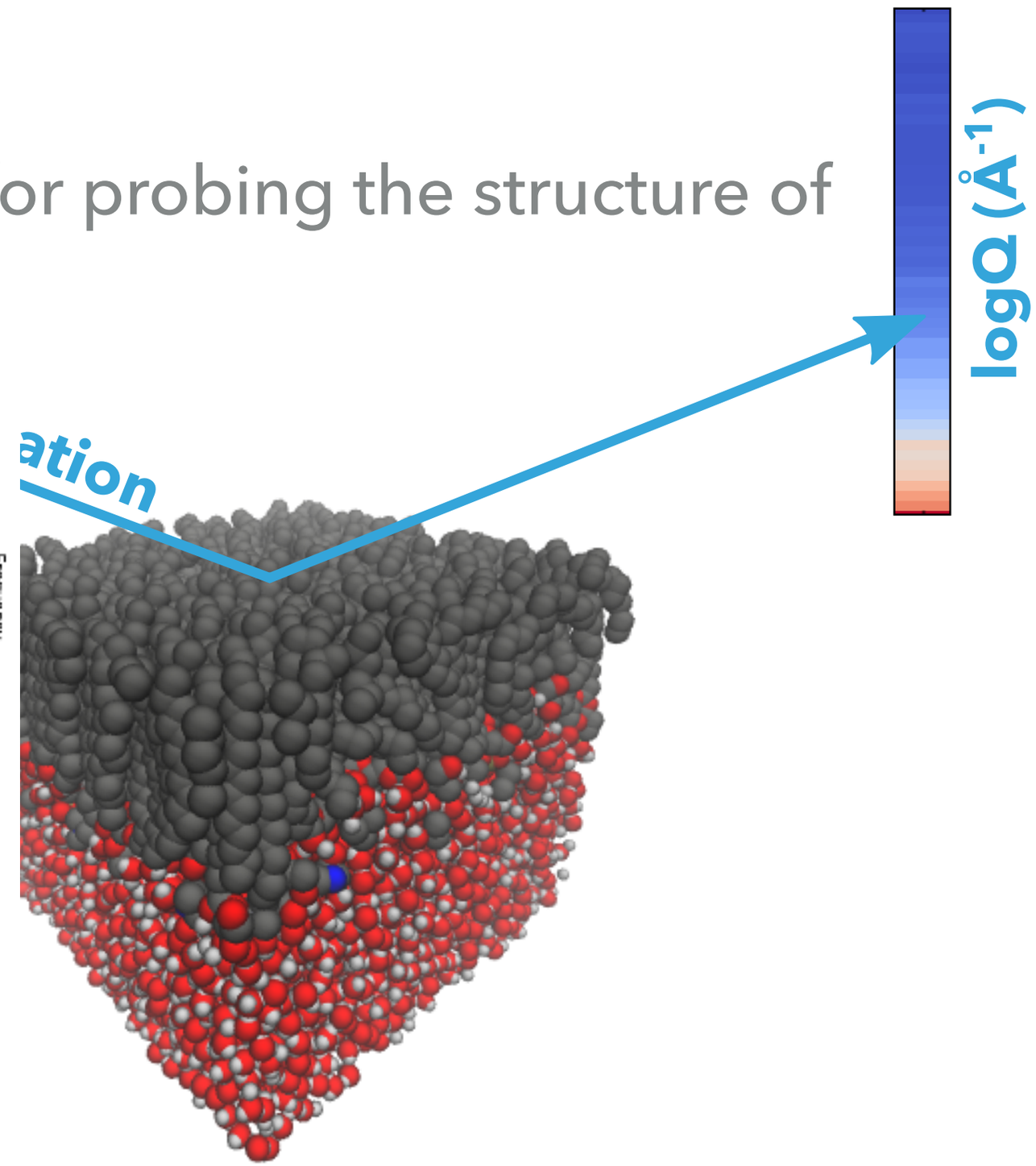



Nanda, H., *et al.* (2010).Biophys. J., 99(8), 2516-2524.



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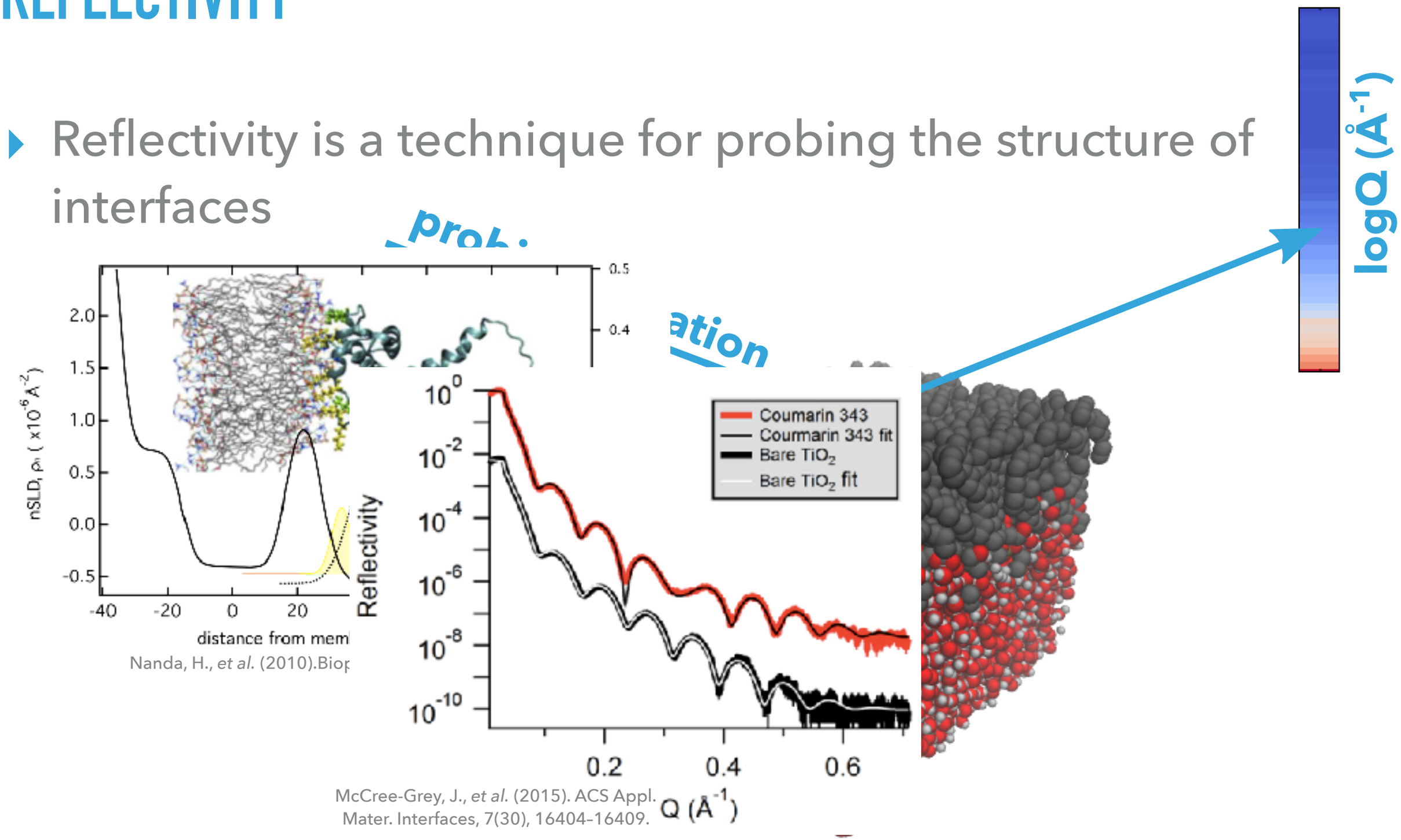


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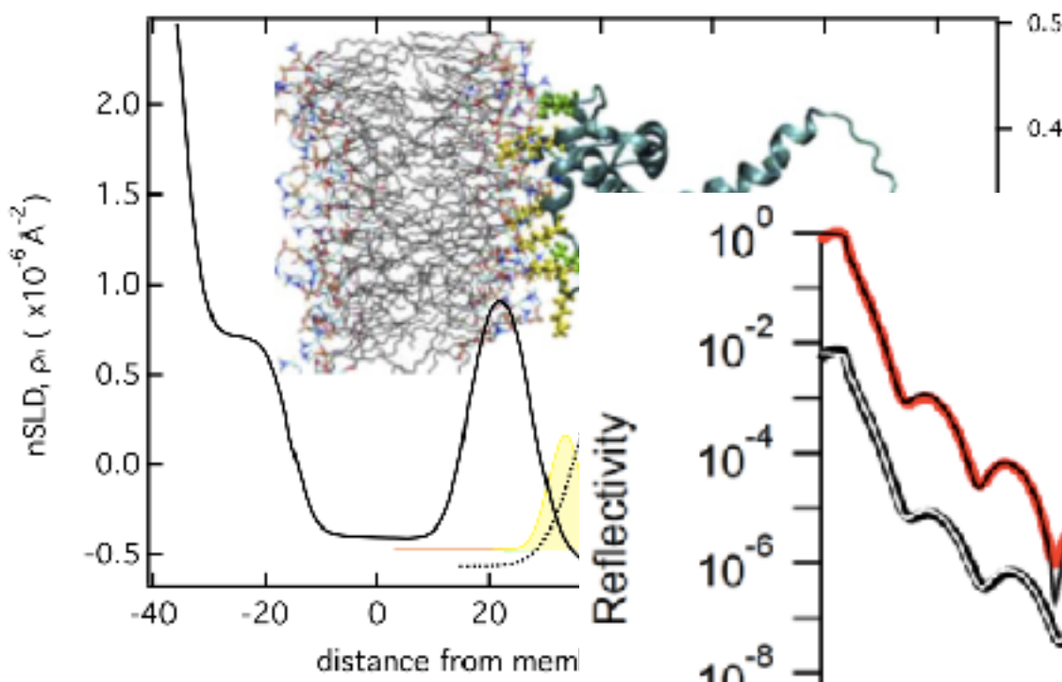


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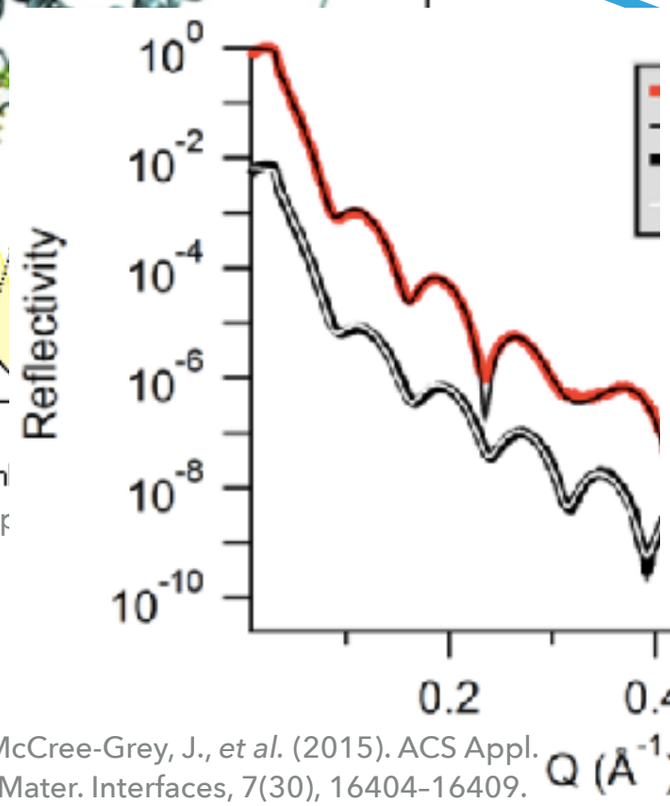
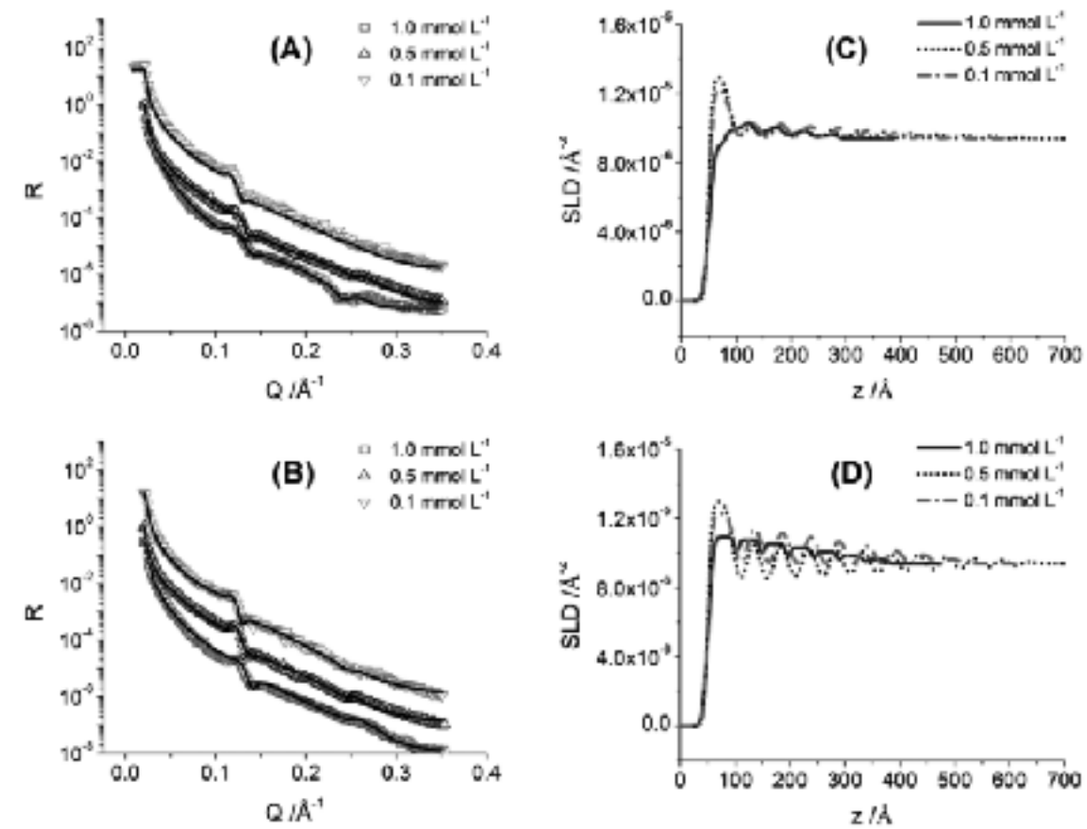



McCree-Grey, J., *et al.* (2015). ACS Appl. Mater. Interfaces, 7(30), 16404–16409.

Comas-Rojas, H., *et al.* (2013). Soft Matter, 9(15), 4003–4014.

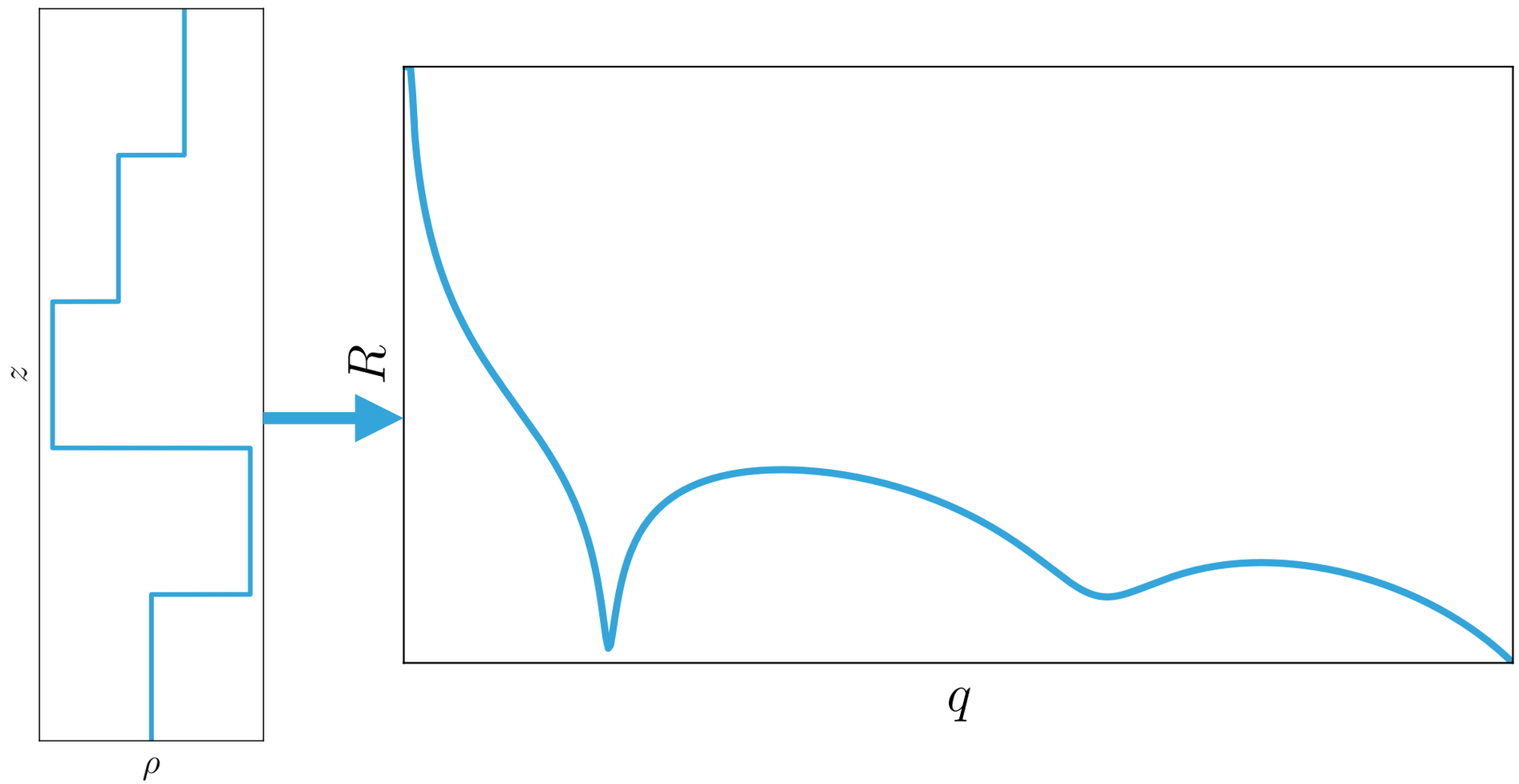


# TRADITIONAL ANALYSIS

- Traditionally reflectivity is analysed with a model of a series of layers of different scattering length density ($\rho$)

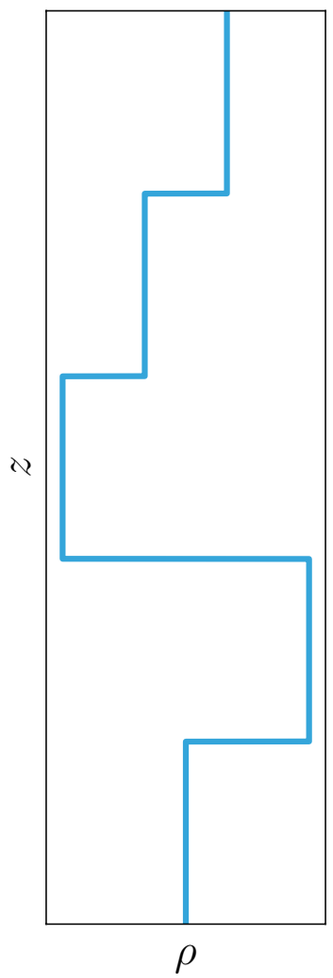
# TRADITIONAL ANALYSIS

- These layers should produce a reflectivity profile which agrees well with the data



# TRADITIONAL ANALYSIS

- Additionally the layers must also have some chemical relevance




THESE LAYERS NEED TO HAVE SOME "CHEMICAL MEANING"

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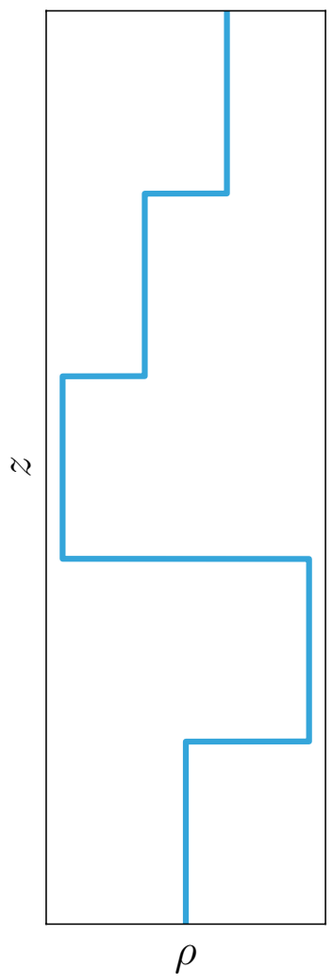
# SIMULATION BASED ANALYSIS

- Instead of trying to determined chemically relevant layers, simulation could be used

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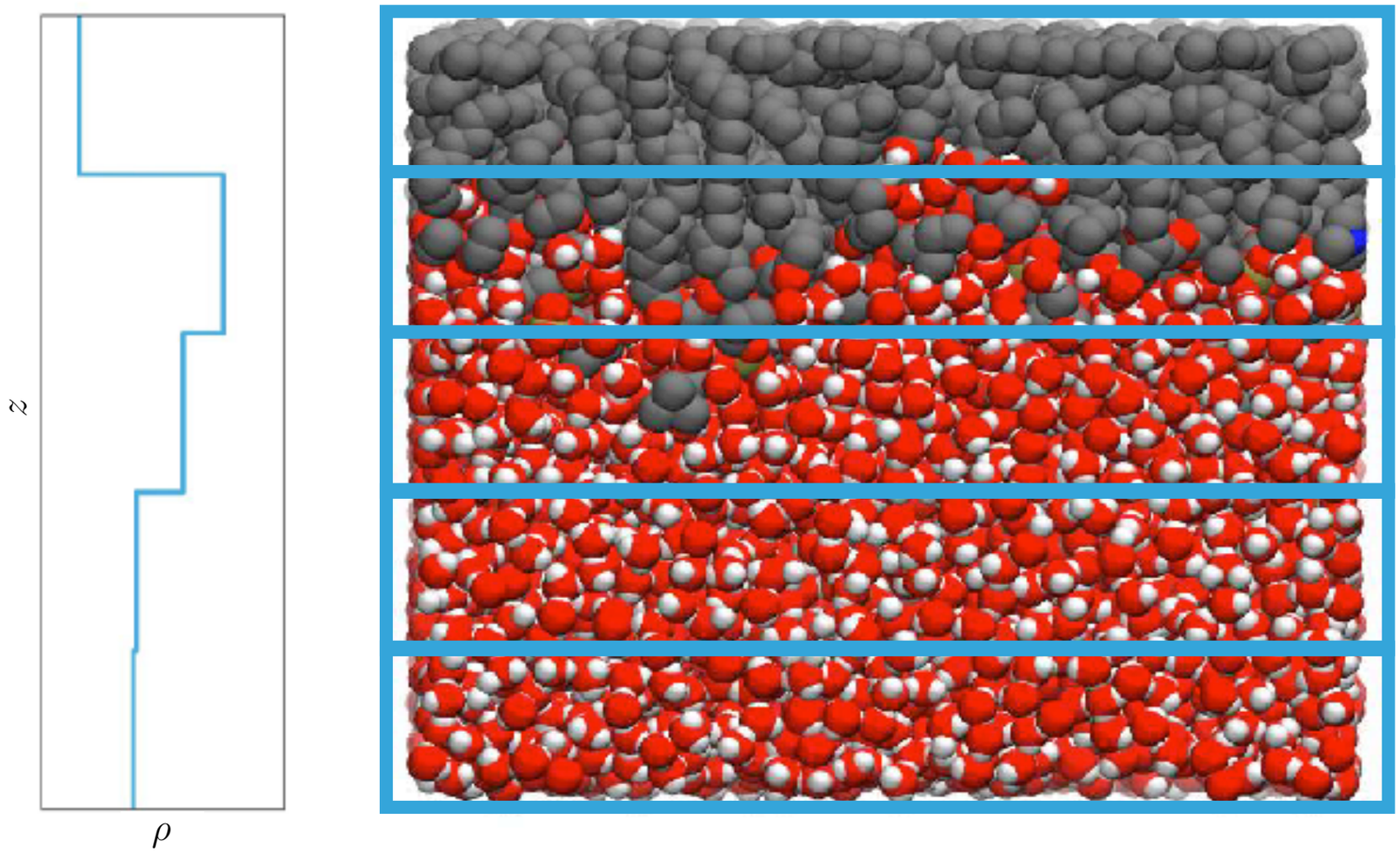
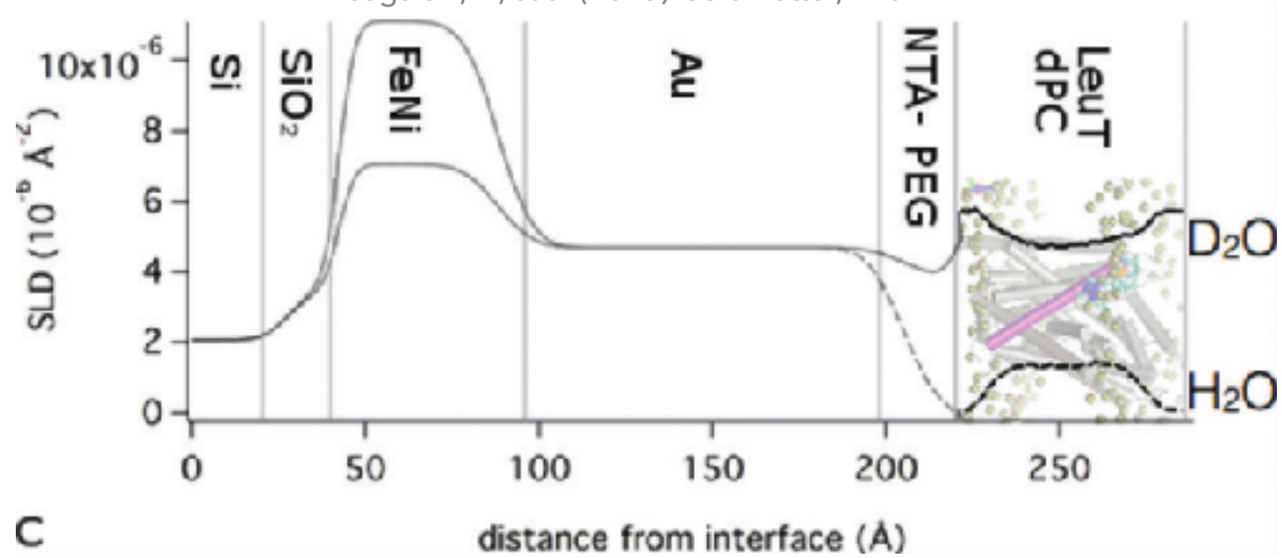
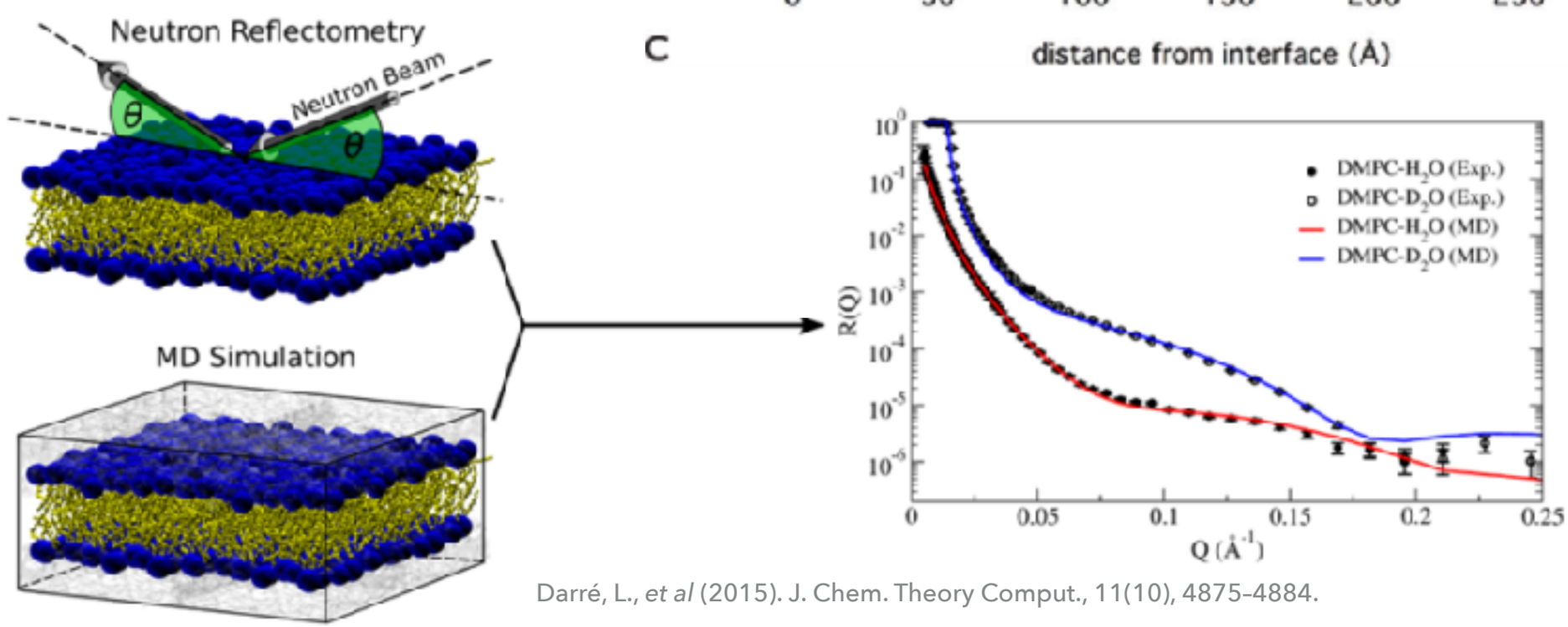


# SIMULATION BASED ANALYSIS

- This method has been applied previously to all-atom systems

Jagalski, V., *et al* (2015). Soft Matter, 7707–7711.



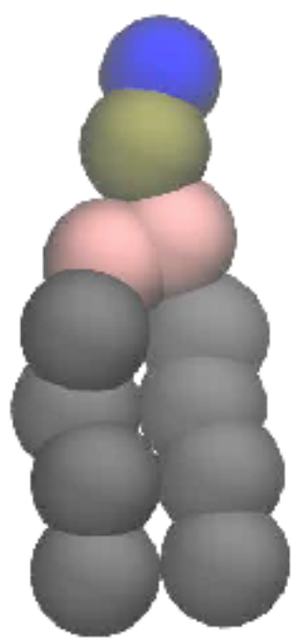

Darré, L., *et al* (2015). J. Chem. Theory Comput., 11(10), 4875–4884.

# SIMULATION BASED ANALYSIS

- Coarse-grained simulations of soft and biological systems are well developed
  - We have studied DSPC monolayers (100 lipids) across 3 forcefield of differing resolution – investigating the effect of the forcefields

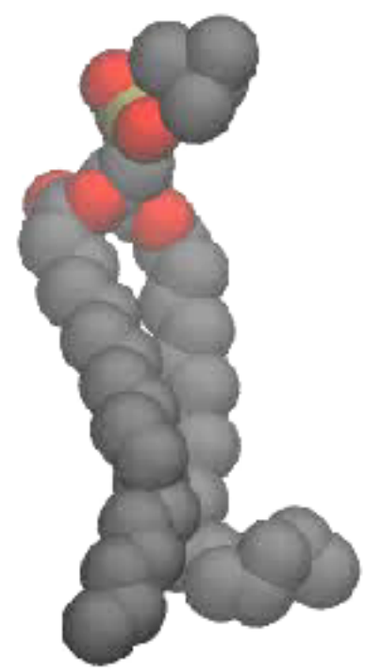
Martini – 4-to-1 CG

*d* = 4 Å



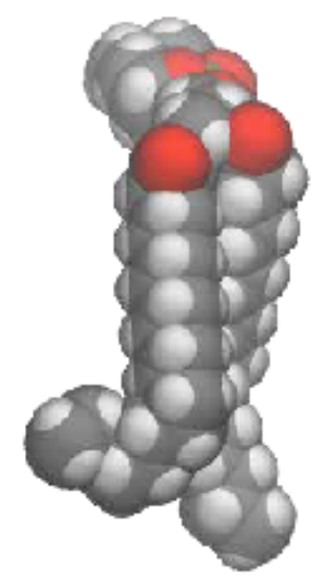
Berger – United Atom

*d* = 2 Å



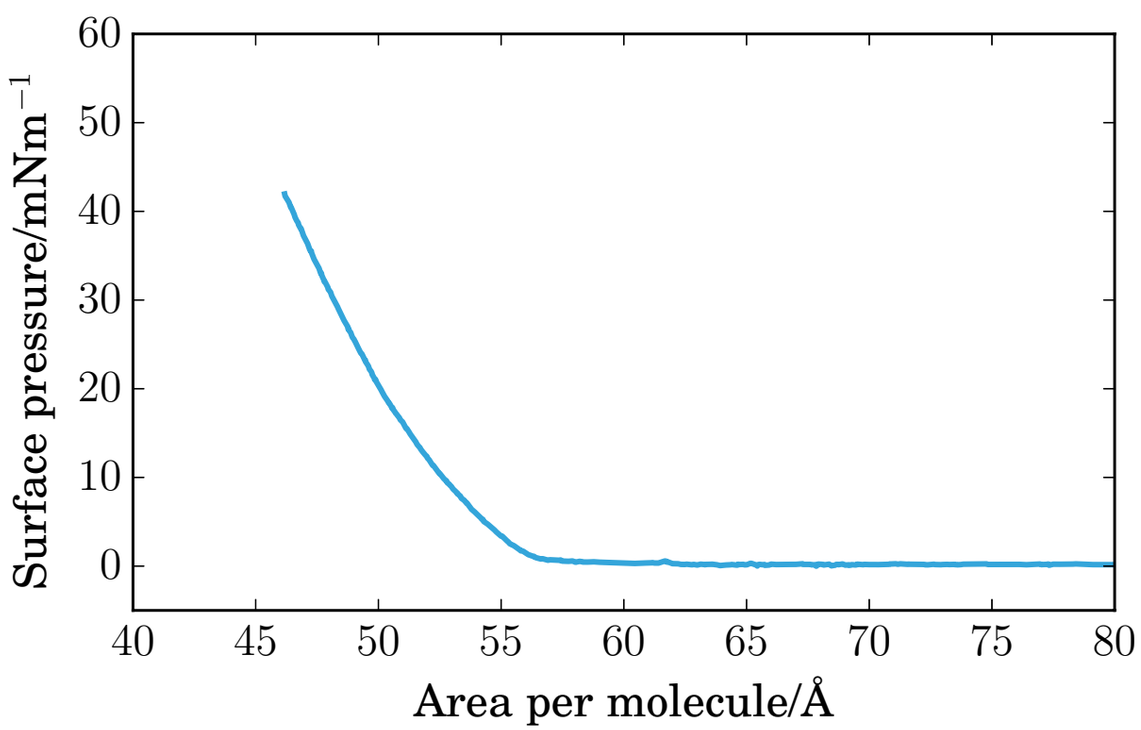
Slipids – All-atom

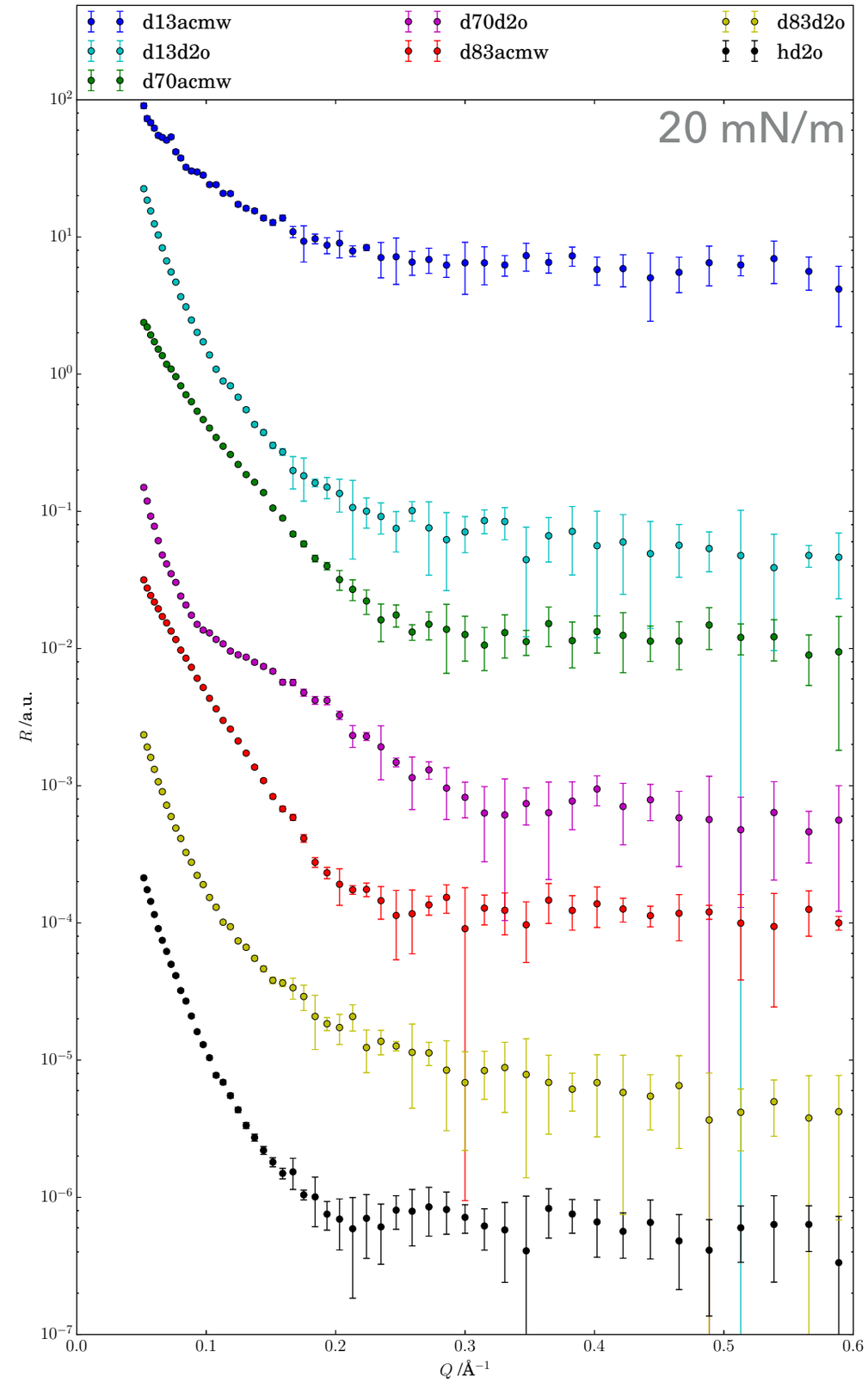
*d* = 1 Å



# EXPERIMENTAL DATA

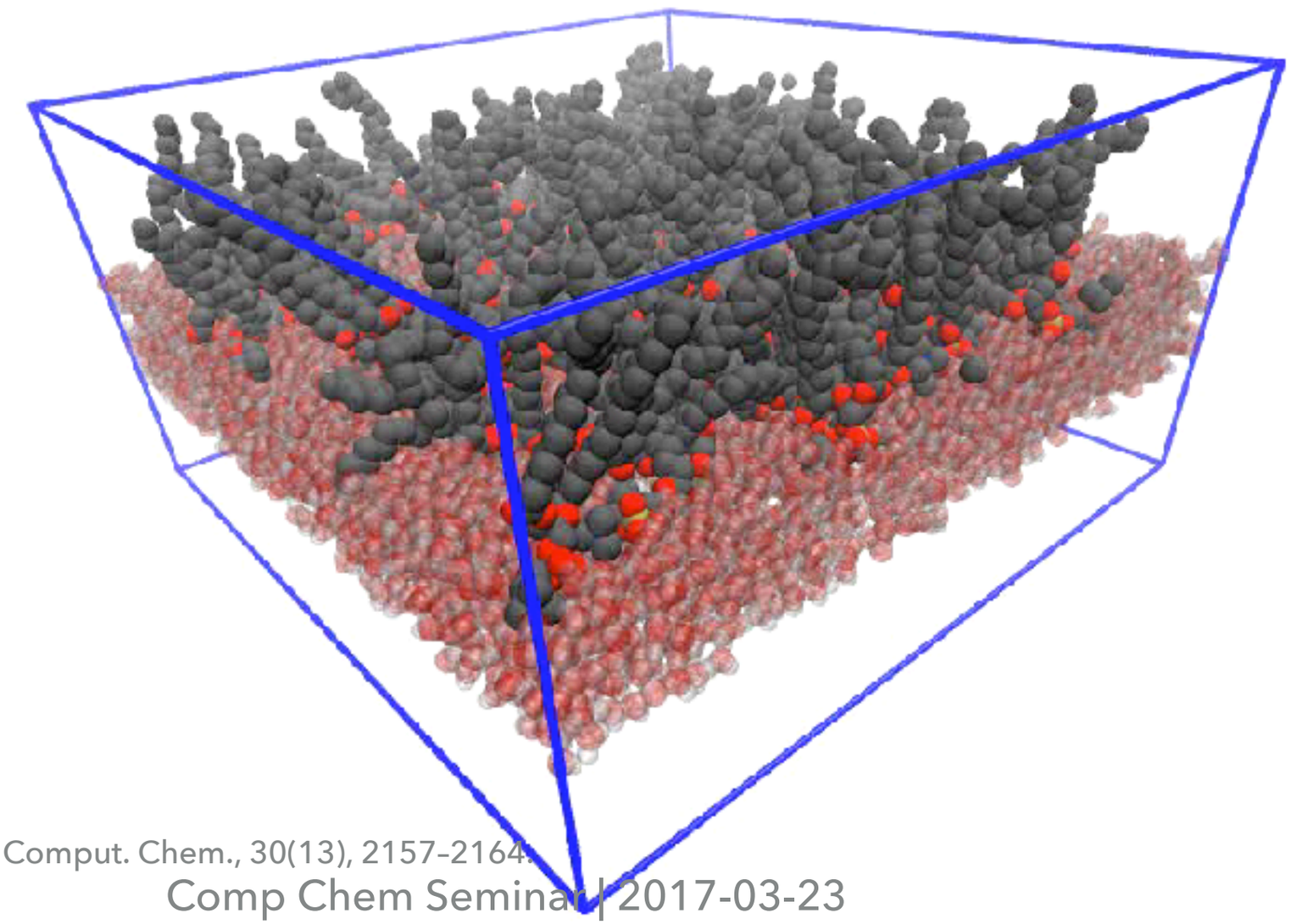
- Neutron reflectivity was collected for the DSPC monolayer at 4 surface pressure (20, 30, 40, 50 mN/m) for 7 isotopic contrasts
  - Experimental surface isotherm was found using traditional method



# MD SIMULATIONS

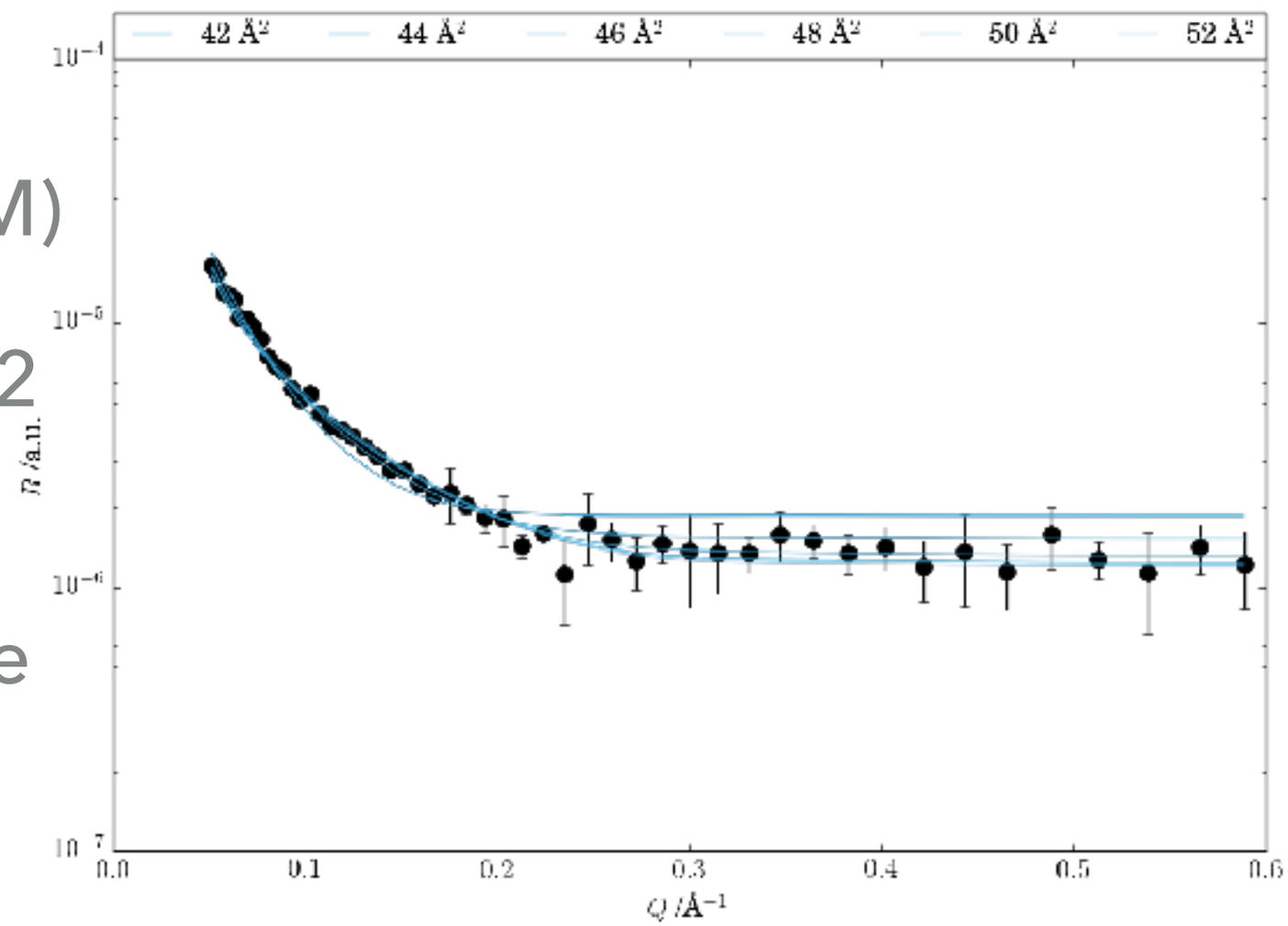
- The lipid monolayer was built with Packmol[1] and subjected to an *in-silico* "Langmuir trough"



[1] Martínez, L., *et al.* (2009). J. Comput. Chem., 30(13), 2157–2164.

# MD SIMULATIONS

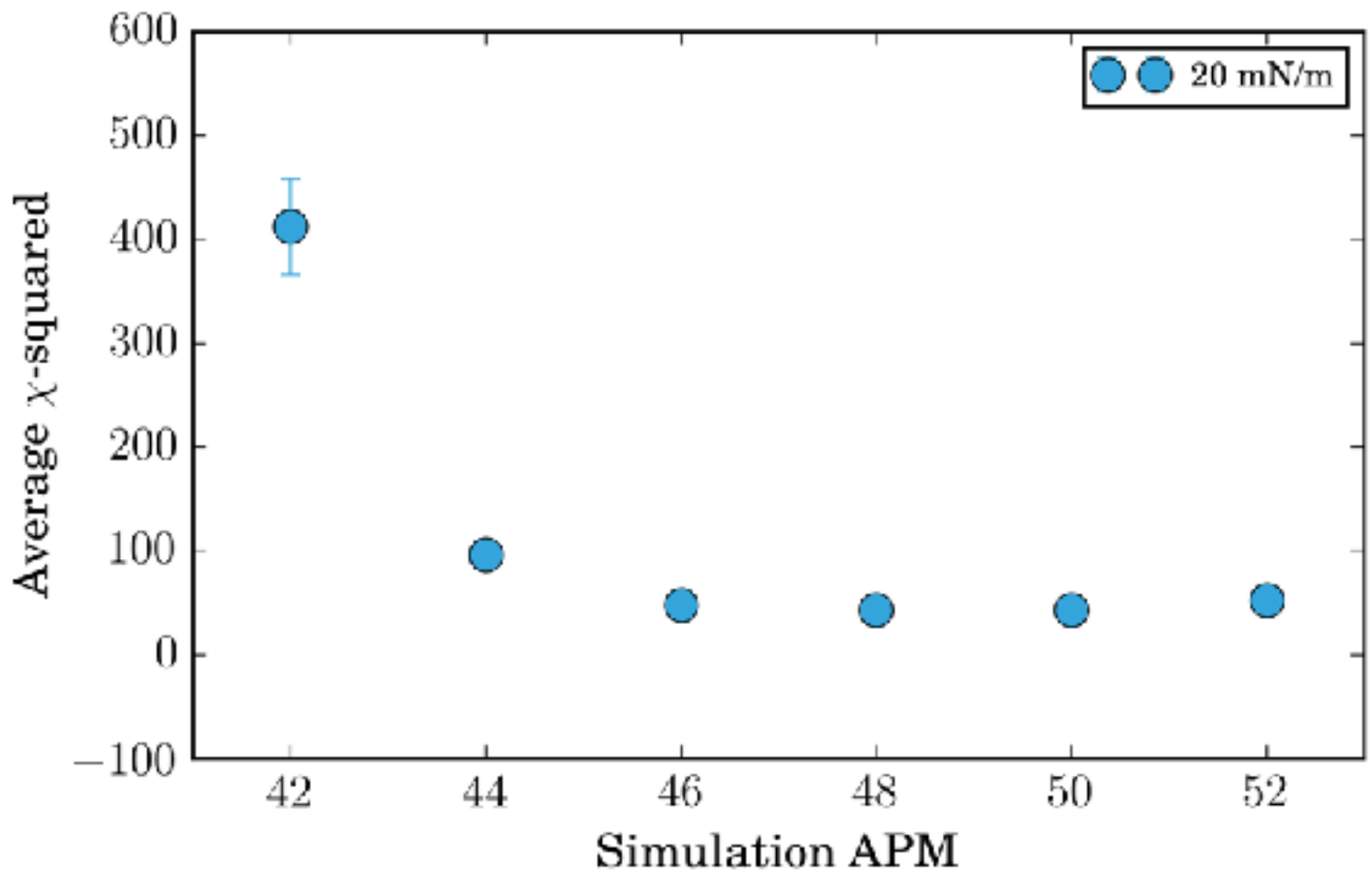
- Each resolution was simulated at a series of areas per molecule (APM)
  - 42, 44, 46, 48, 50, & 52 plus 40 for Martini
- The reflectivity each time was found using falass




falass, www.people.bath.ac.uk/chske/research/software/falass/index.html

# SIMULATED SURFACE ISOTHERM

- Each simulation was then compared with the experimental data for each surface pressure at all contrasts and the average goodness-of-fit ($\chi^2$) found



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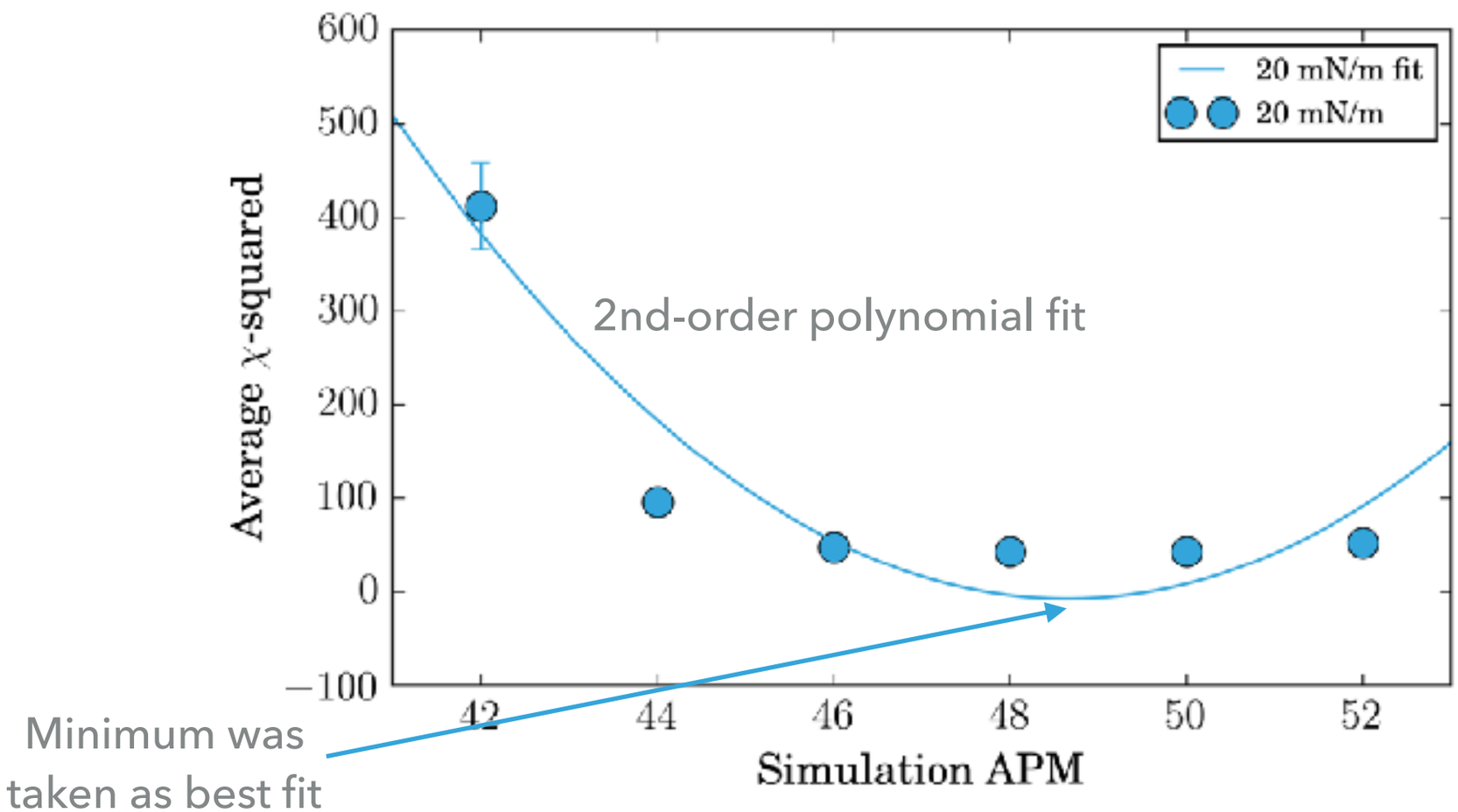

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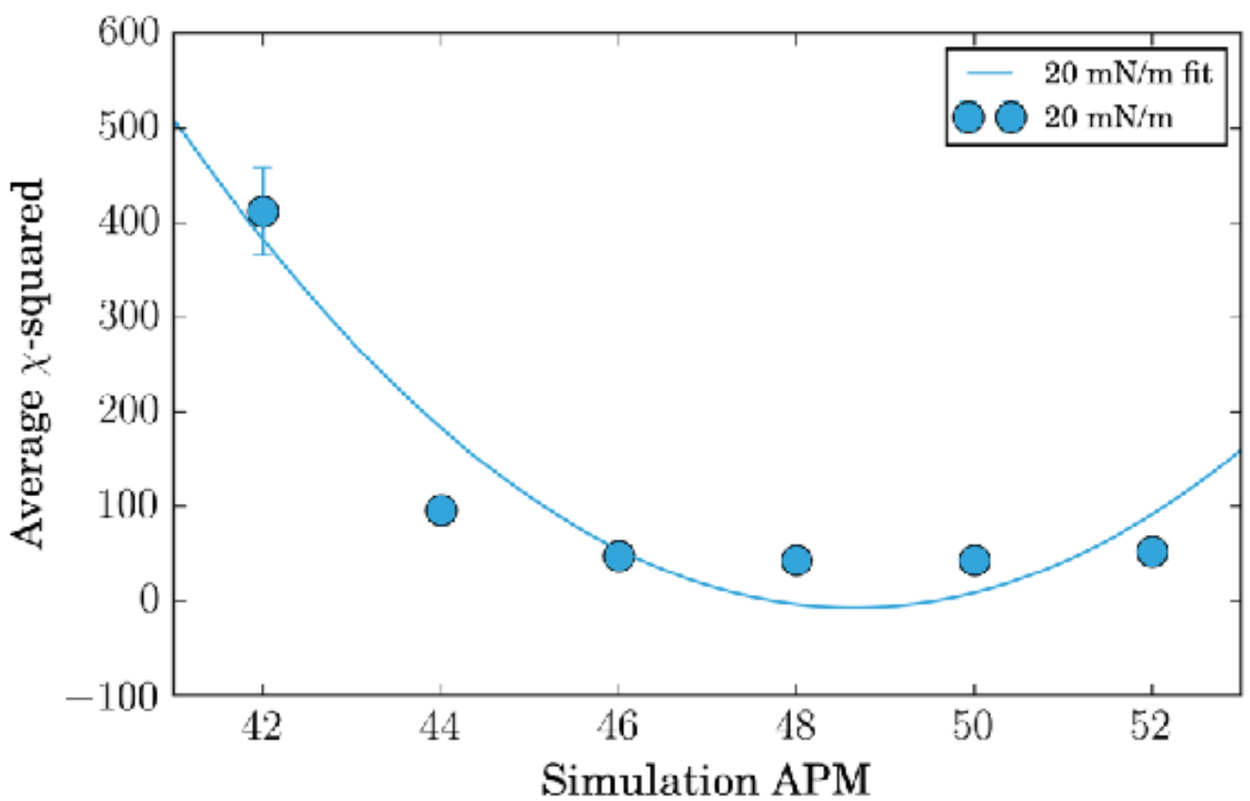
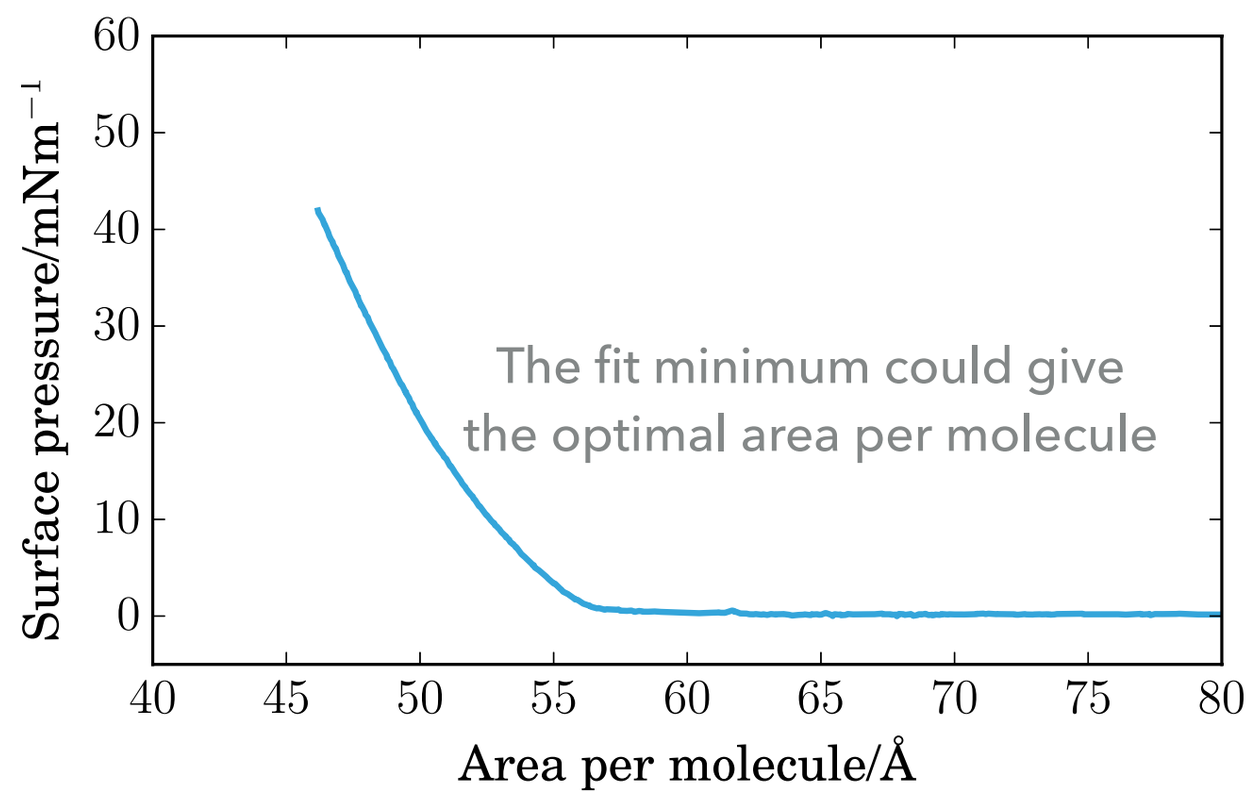
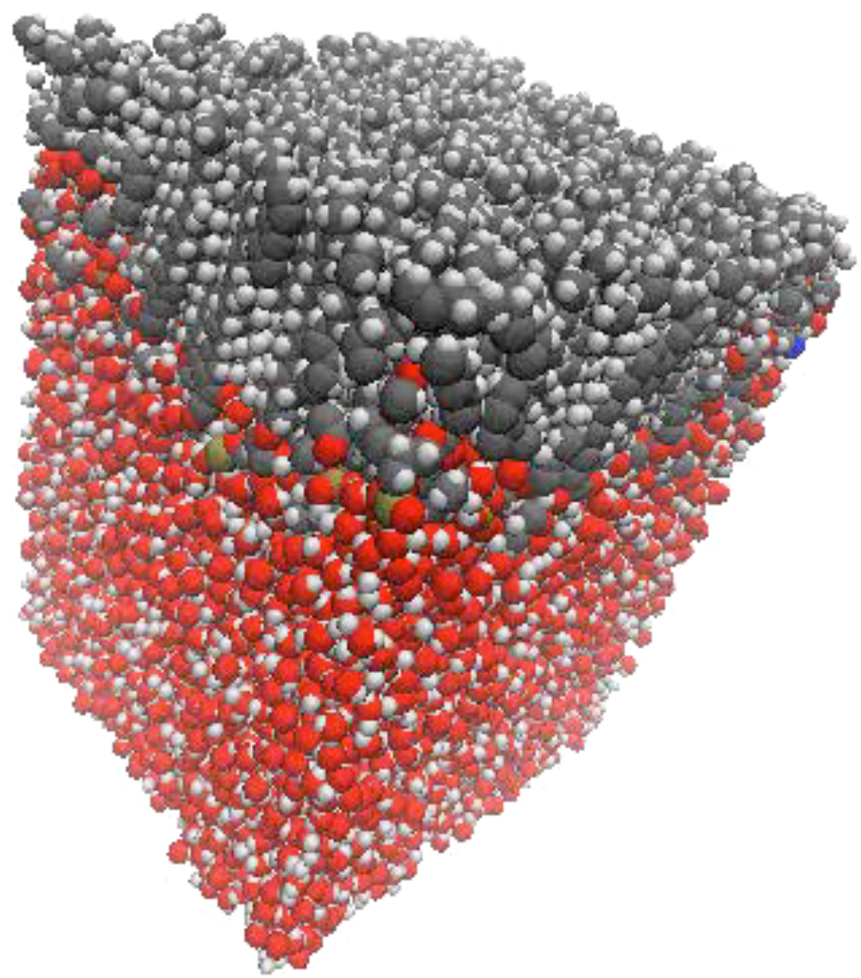



The fit minimum could give the optimal area per molecule

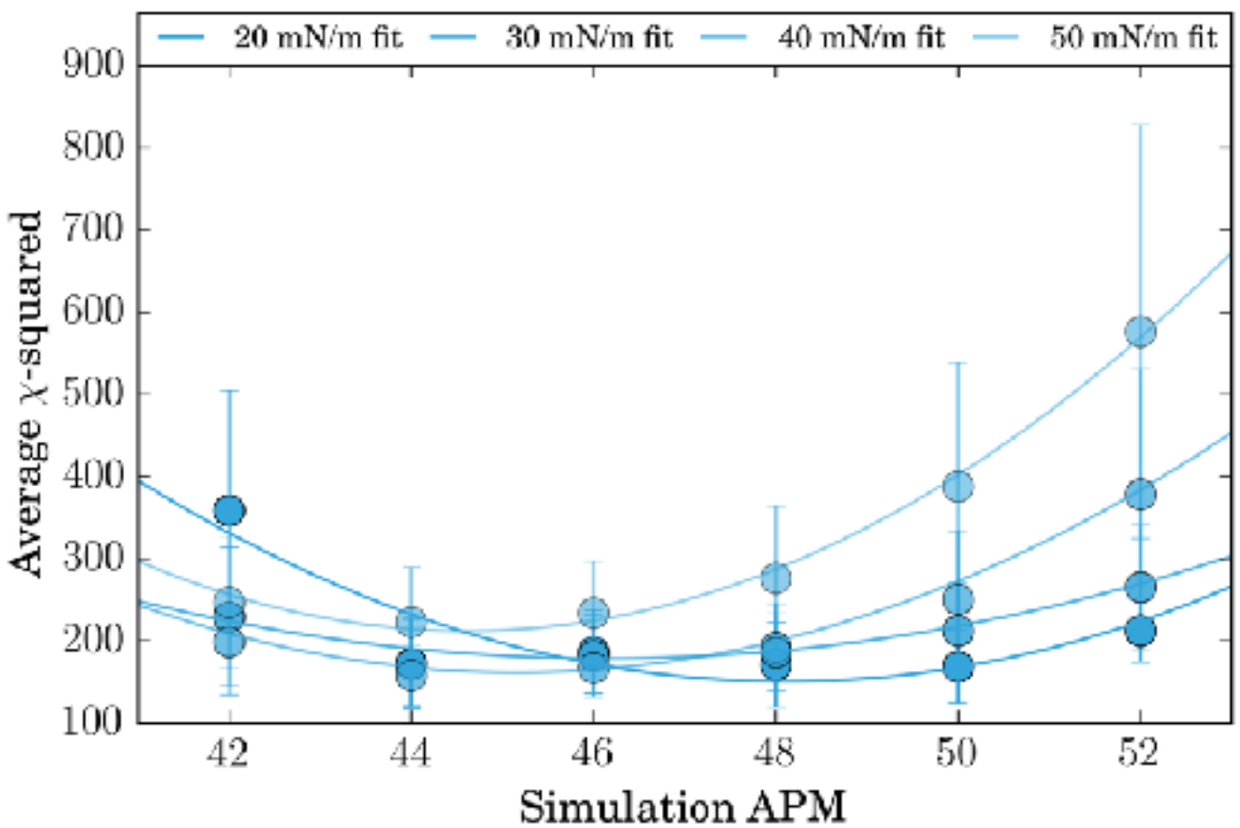
# SLIPID

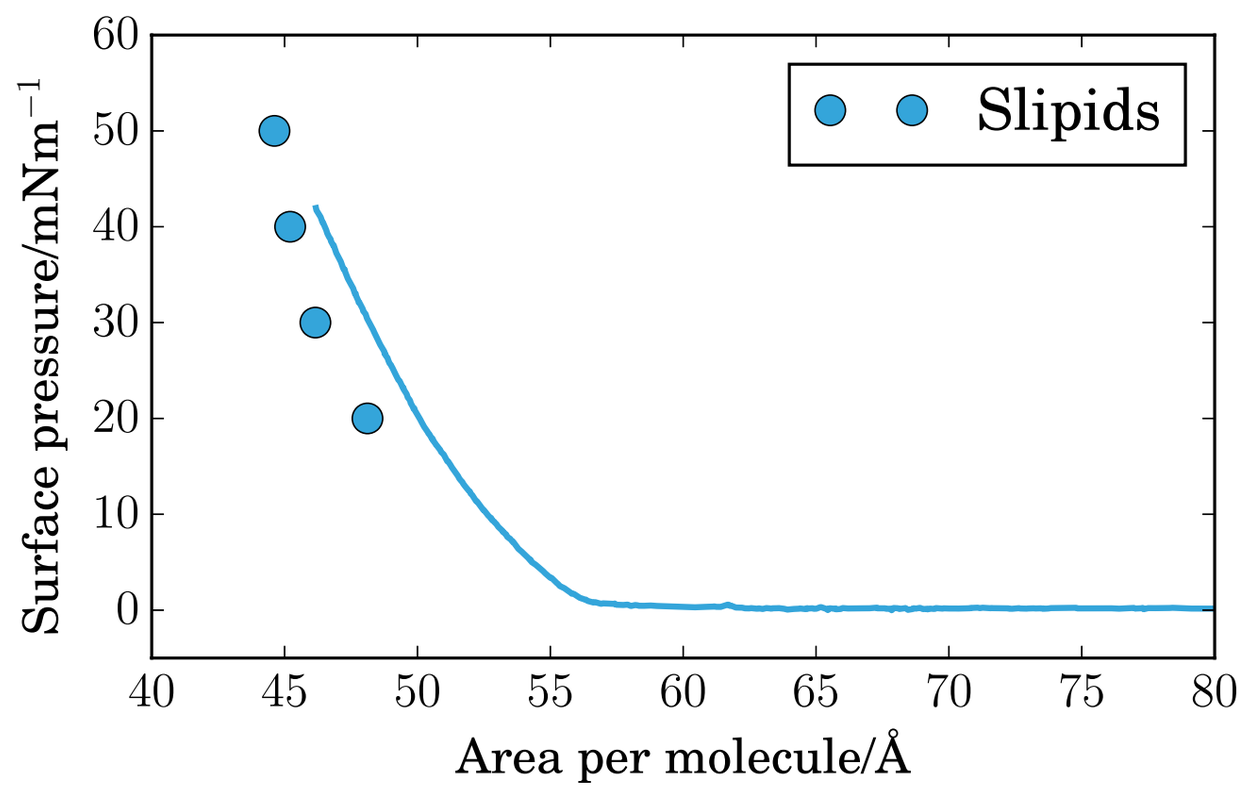
- The all-atom lipid simulations took ~ 150 hours of core time on SCARF
  - $\Delta t$ = 0.5 fs



# SLIPID

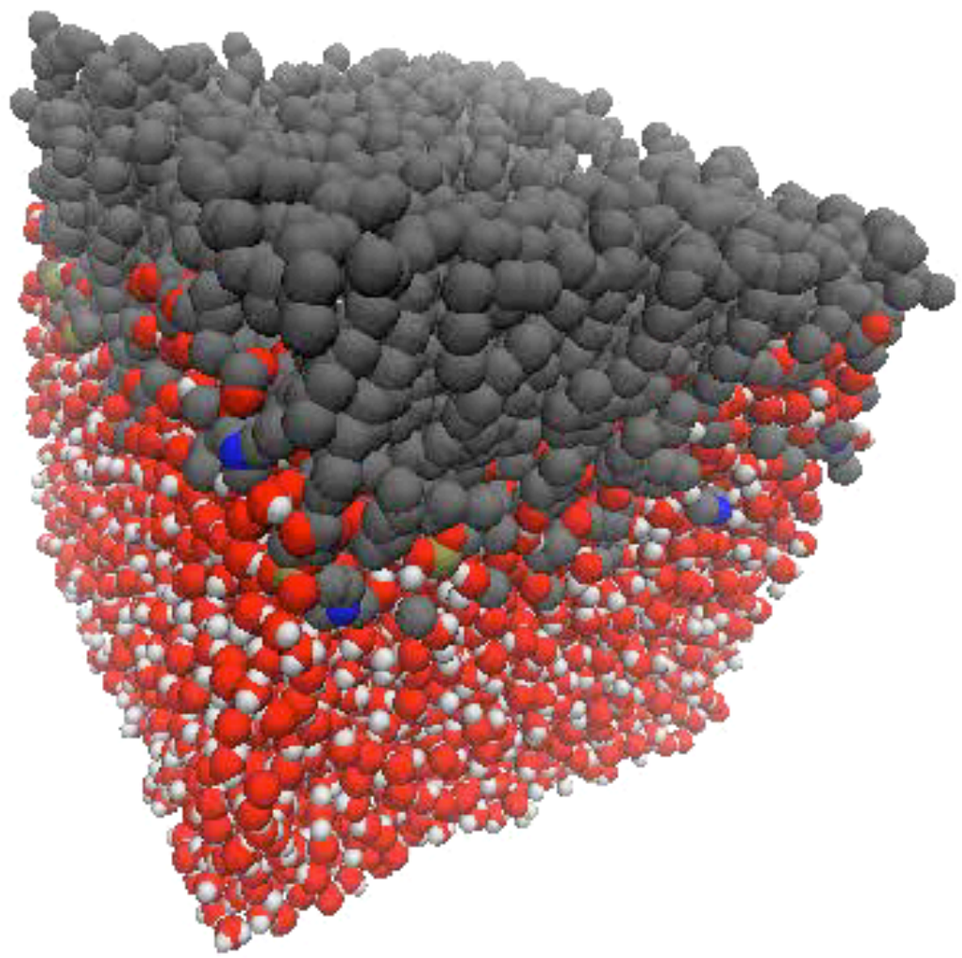
- The all-atom simulations gave rise to the following simulated surface isotherm



# BERGER

- The united-atom lipid simulations took ~ 60 hours of core time on SCARF
  - $\Delta t$ = 2 fs



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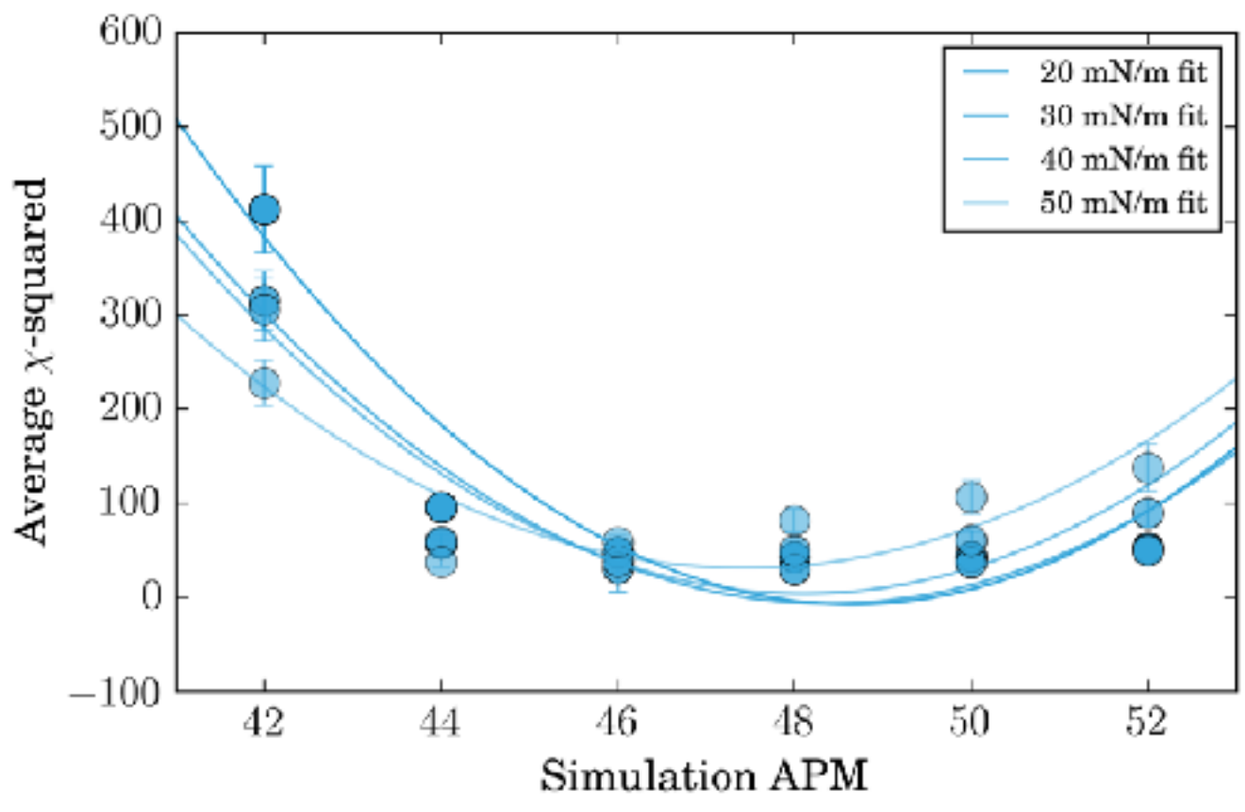
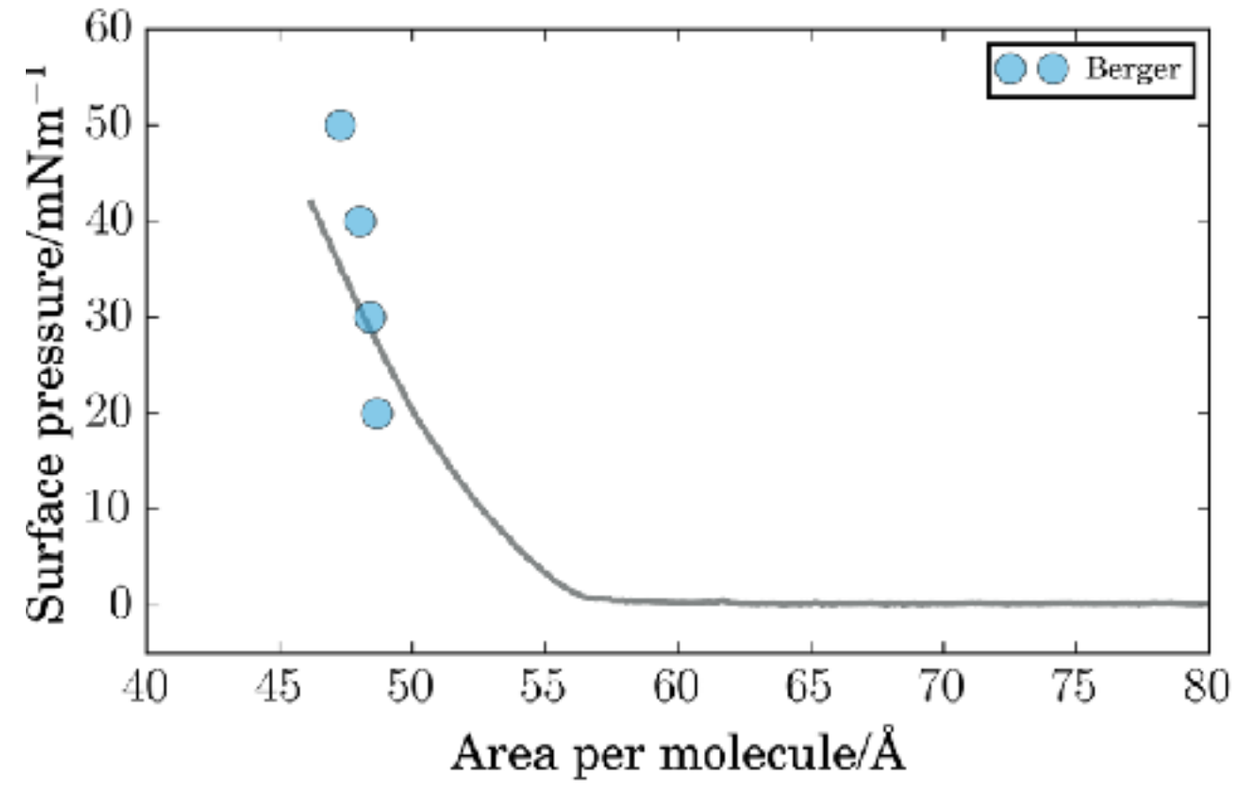
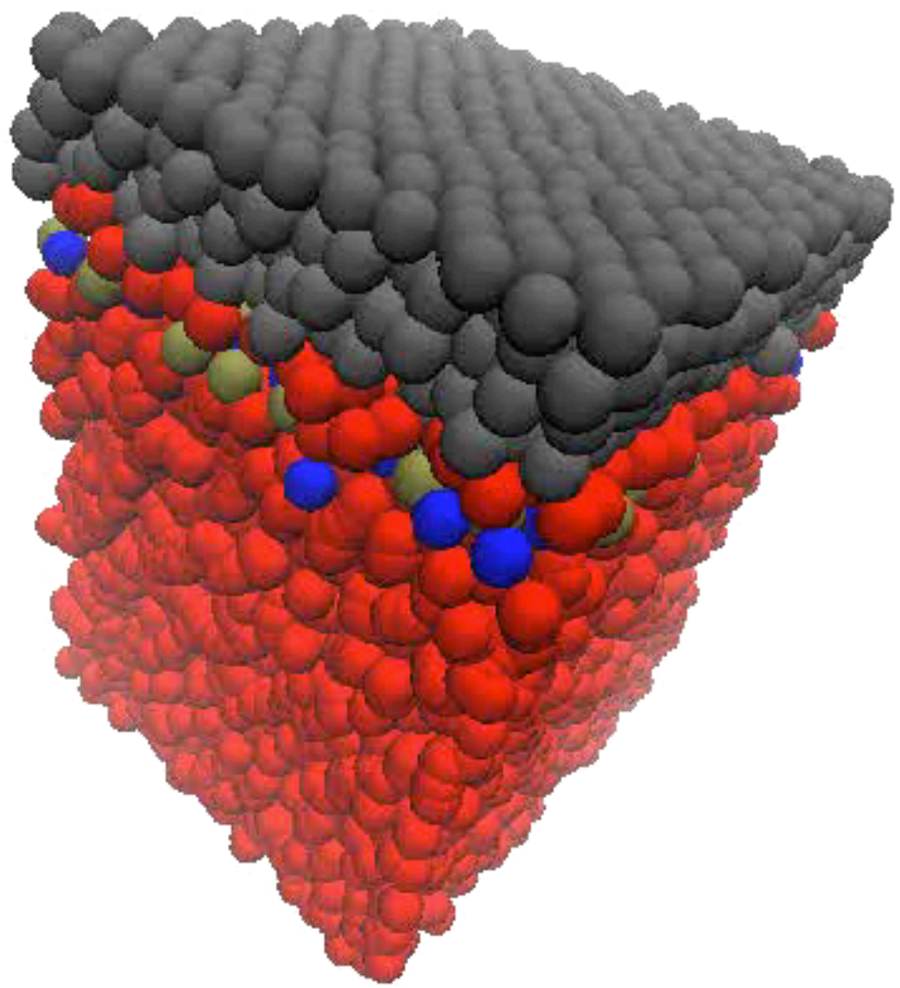


# MARTINI

- The coarse-grained lipid simulations took ~ 38 hours of core time on SCARF
  - $\Delta t$ = 20 fs



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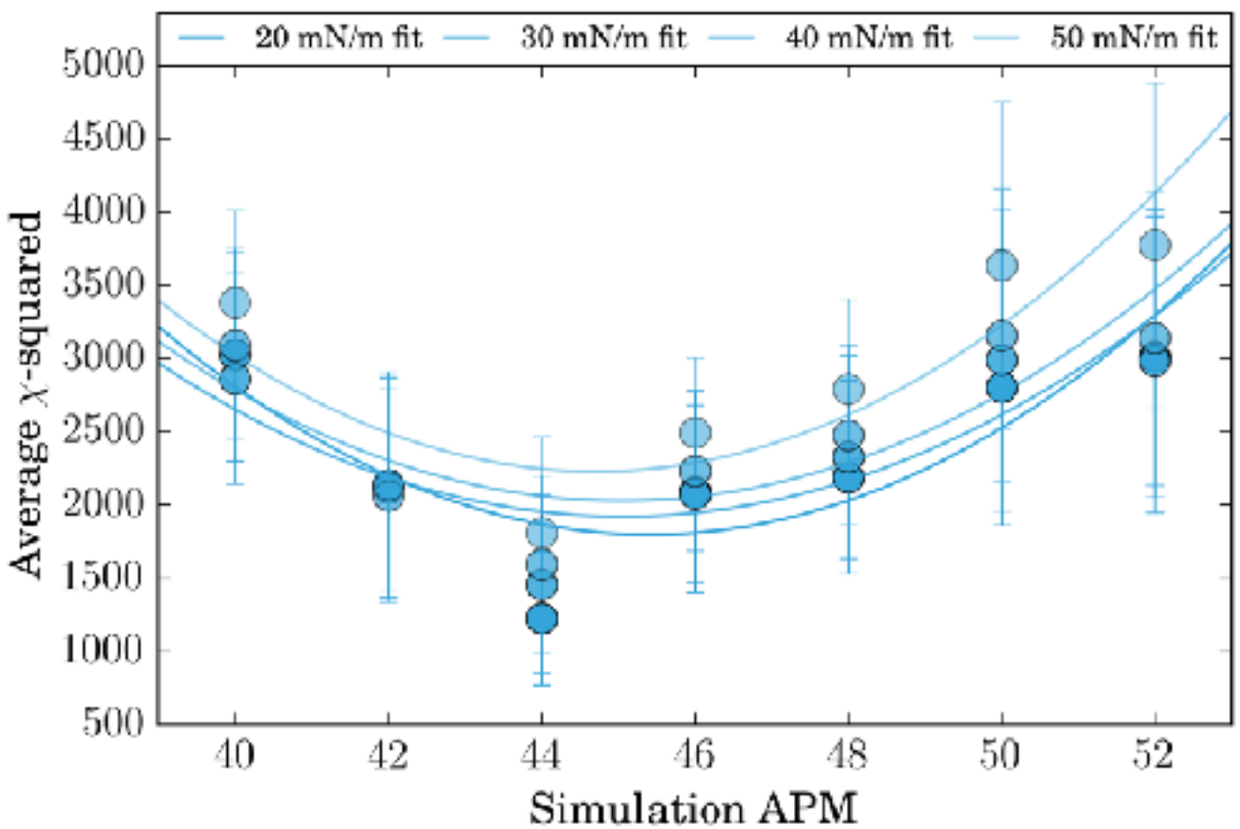
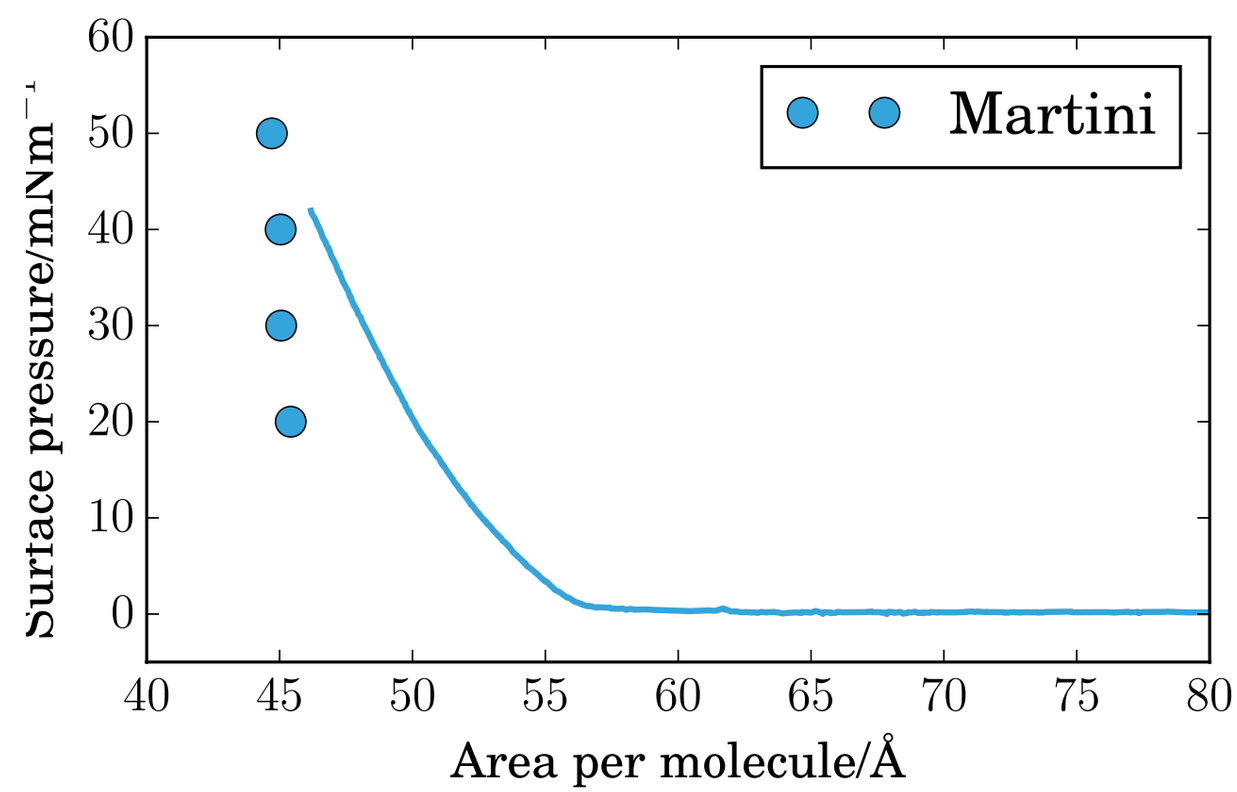
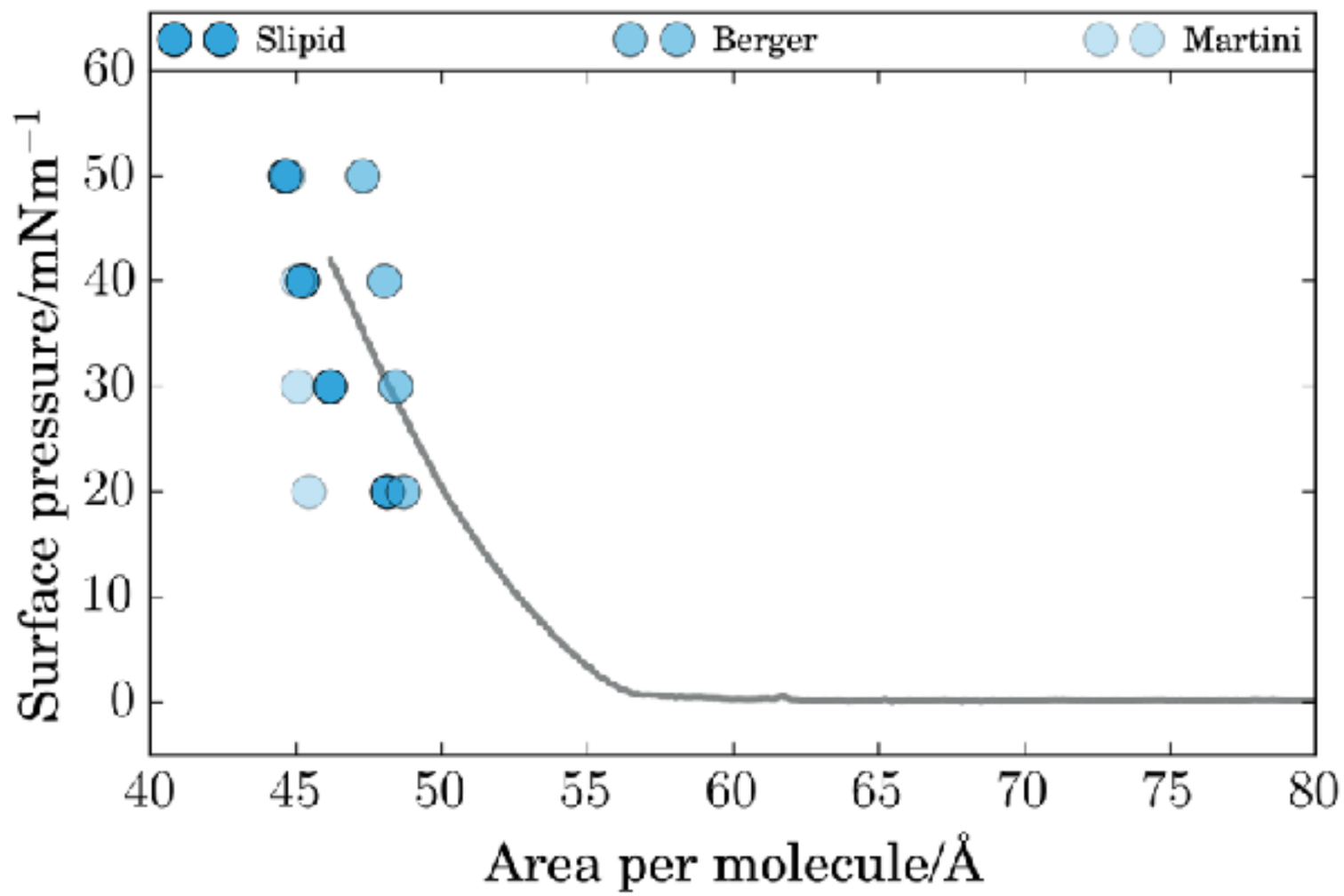


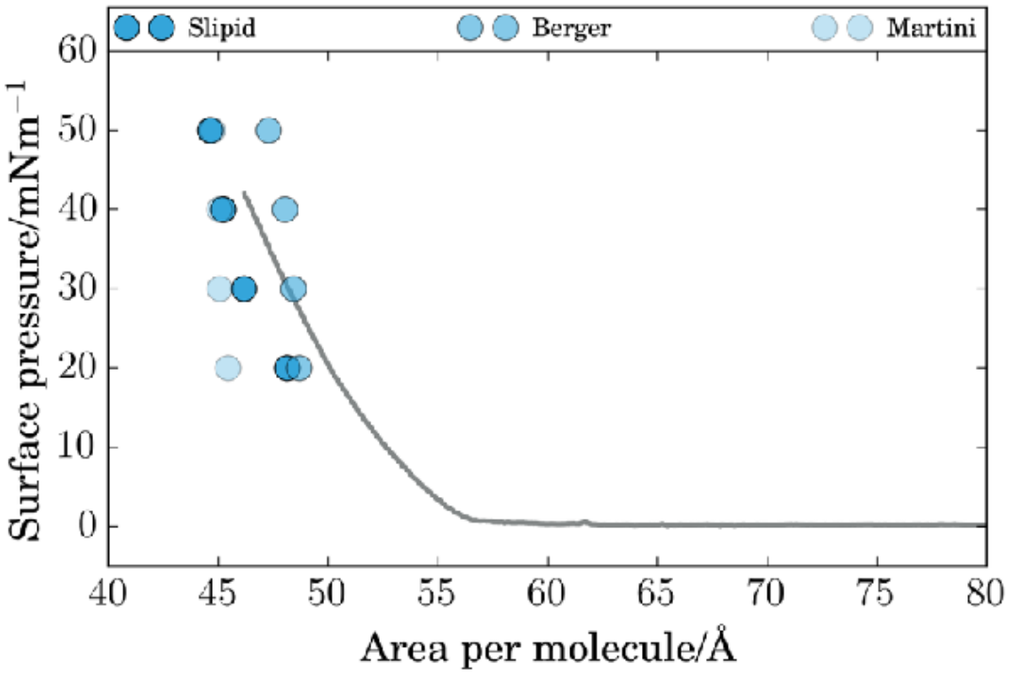
# COMPARISON

- The three resolutions are compared with the traditional method for determining APM
  - Slipids has the right shape, although a bit offset
  - Berger is not offset, however too steep
  - Martini is offset and too steep



# CONCLUSIONS

- Either Slipids or the Berger potentials are best for comparison with reflectivity measurements
  - Use of Berger potentials allow a two times simulation speed up
  - Slipids potentials more accurately modelling the compressibility of the monolayer
  - Although Martini is not a accurate as the others – offers a significant speed up



# THANKS


- Acknowledgements
  - Karen Edler (Bath)
  - Steve Parker (Bath)
  - James Grant (Bath)
  - Rob Barker (Dundee)
  - Bath/DLS – Funding

Sadie